

```
/* CMP-135 SDTM EX Creation
Domain: EX*/;
options validvarname=upcase nofmterr dlcreatedir;

libname raw  "/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1";
libname sdtm  "/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm";

libname valxpt xport
"/home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/ex.xpt";

libname sexpt xport
"/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1/sdtm to use for vs ds/se.xpt";
```

```
data se_prep;
    set sexpt.se;

    sestdt = input(substr(SESTDTC, 1, 10), yymmdd10.);
    seendt = input(substr(SEENDTC, 1, 10), yymmdd10.);

    format sestdt seendt date9.;

    keep USUBJID EPOCH sestdt seendt;
run;
```

```
data ex_pre;
    length
        STUDYID  $6
        DOMAIN   $2
        EXTRT    $10
        EXDOSU   $2
        EXDOSFRM $9
        EXROUTE  $12
        EXLOC2   $20
        EXSTDTC  $16
        USUBJID  $40;

    set raw.ex;

    STUDYID = "CMP135";
    DOMAIN  = "EX";
    USUBJID = catx("-", STUDYID, strip(SUBJECT));

    EXTRT    = "CMP135";
    EXDOSE2  = EXDOSE;
    EXDOSU   = "mg";
    EXDOSFRM = "INJECTION";
    EXROUTE  = "SUBCUTANEOUS";

    if not missing(EXLOCOTH) then EXLOC2 = upcase(strip(EXLOCOTH));
    else EXLOC2 = upcase(strip(EXLOC));

    if vtype(EXSTDTCN) = "N" then do;
        if not missing(EXSTDTCN) then do;
```

```

        if abs(EXSTDTN) > 50000 then EXSTDT = datepart(EXSTDTN);
        else EXSTDT = EXSTDTN;
    end;
end;
else do;
    EXDTM_RAW = input(strip(EXSTDTN), anydtdtm.);
    if not missing(EXDTM_RAW) then EXSTDT = datepart(EXDTM_RAW);
    if missing(EXSTDT) then EXSTDT = input(strip(EXSTDTN), anydtdte.);
end;

if vtype(EXSTTM) = "N" then do;
    if EXSTTM > 86400 then EXSTM = timepart(EXSTTM);
    else EXSTM = EXSTTM;
end;
else EXSTM = input(strip(EXSTTM), time.);

if not missing(EXSTDT) and not missing(EXSTM) then
    EXSTDTC = cats(put(EXSTDT, yymmdd10.), "T", put(EXSTM, tod5.));
else if not missing(EXSTDT) then
    EXSTDTC = put(EXSTDT, yymmdd10.);

format EXSTDT date9. EXSTM time8.;

keep STUDYID DOMAIN USUBJID EXTRT EXDOSE2 EXDOSU EXDOSFRM
    EXROUTE EXLOC2 EXSTDTC EXSTDT SUBJECT RECORDID RECORDPOSITION;

run;

```

```

proc sql;
    create table ex_rfstdt as
    select
        SUBJECT,
        min(EXSTDT) as RFSTDT format=date9.
    from ex_pre
    where not missing(EXSTDT)
    group by SUBJECT;
quit;

```

```

proc sql;
    create table ex_with_dy as
    select
        a.*,
        b.RFSTDT
    from ex_pre as a
    left join ex_rfstdt as b
        on a.SUBJECT = b.SUBJECT;
quit;

```

```

data ex_with_dy;
    set ex_with_dy;

    if not missing(EXSTDT) and not missing(RFSTDT) then do;
        if EXSTDT >= RFSTDT then EXSTDY = EXSTDT - RFSTDT + 1;
        else EXSTDY = EXSTDT - RFSTDT;
    end;
run;

```

```
proc sql;
  create table ex_epoch_match as
  select
    a.*,
    b.EPOCH,
    b.SEESTDT,
    b.SEENDT
  from ex_with_dy as a
  left join se_prep as b
    on a.USUBJID = b.USUBJID
    and a.EXSTDTC >= b.SEESTDT
    and a.EXSTDTC <= b.SEENDT;
quit;
```

```
proc sort data=ex_epoch_match;
  by STUDYID USUBJID RECORDID RECORDPOSITION descending SEESTDT;
run;
```

```
data ex_with_epoch;
  set ex_epoch_match;
  by STUDYID USUBJID RECORDID RECORDPOSITION;

  if first.RECORDPOSITION;

  drop SEESTDT SEENDT RFSTDTC;
run;
```

```
proc sort data=ex_with_epoch out=ex_sorted;
  by STUDYID USUBJID EXSTDTC EXSTDTC RECORDPOSITION;
run;
```

```
data sdtm.ex;
  retain
    STUDYID DOMAIN EXSEQ EXTRT EXDOSE EXDOSU EXDOSFRM
    EXROUTE EXLOC EPOCH EXSTDTC EXSTDY USUBJID;

  length
    STUDYID $6
    DOMAIN $2
    EXTRT $10
    EXDOSU $2
    EXDOSFRM $9
    EXROUTE $12
    EXLOC $20
    EPOCH $11
    EXSTDTC $16
    USUBJID $40;

  set ex_sorted;
  by STUDYID USUBJID;

  if first.USUBJID then EXSEQ = 1;
  else EXSEQ + 1;
```

```
EXDOSE = EXDOSE2;
EXLOC  = EXLOC2;
```

```
label
```

```
STUDYID = "Study Identifier"
DOMAIN  = "Domain Abbreviation"
EXSEQ    = "Sequence Number"
EXTRT    = "Name of Actual Treatment"
EXDOSE   = "Dose per Administration"
EXDOSU   = "Dose Units"
EXDOSFRM = "Dose Form"
EXROUTE  = "Route of Administration"
EXLOC    = "Location of Dose Administration"
EPOCH    = "Epoch"
EXSTDTC  = "Start Date/Time of Treatment"
EXSTDY   = "Study Day of Start of Treatment";
```

```
keep
```

```
STUDYID DOMAIN EXSEQ EXTRT EXDOSE EXDOSU EXDOSFRM
EXROUTE EXLOC EPOCH EXSTDTC EXSTDY USUBJID;
```

```
run;
```

```
proc sort data=sdm.ex out=ex_created;
  by STUDYID USUBJID EXSEQ;
run;
```

```
proc sort data=valxpt.ex out=ex_validate;
  by STUDYID USUBJID EXSEQ;
run;
```

```
ods rtf file="/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/EX_PROC_COMPARE.rtf";
```

```
title "PROC COMPARE: SDTM EX Created vs Validation EX";
```

```
proc compare base=ex_validate compare=ex_created listall criterion=0.0000001;
  id STUDYID USUBJID EXSEQ;
run;
```

```
ods rtf close;
```

```
title "Created EX - First 5 Records";
```